

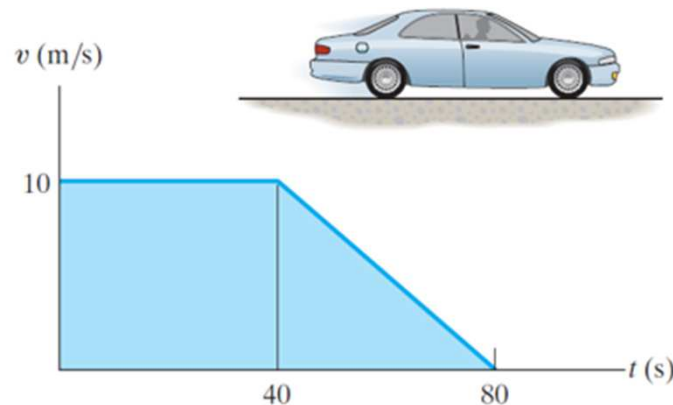
동역학

(Engineering Mechanics: Dynamics)

Chapter 12-2

영남대학교 기계공학부 김병일 교수

Group Problem Solving I: V-T Graph (1)



Given: The v-t graph shown.

Find: The a-t graph, average speed, and distance traveled for the 0 - 80 s interval.

Plan: Find slopes of the v-t curve and draw the a-t graph.

Find the area under the curve. It is the distance traveled.

Finally, calculate average speed (using basic definitions!).



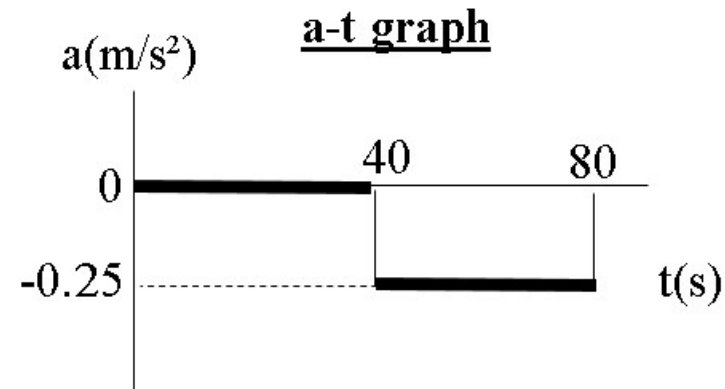
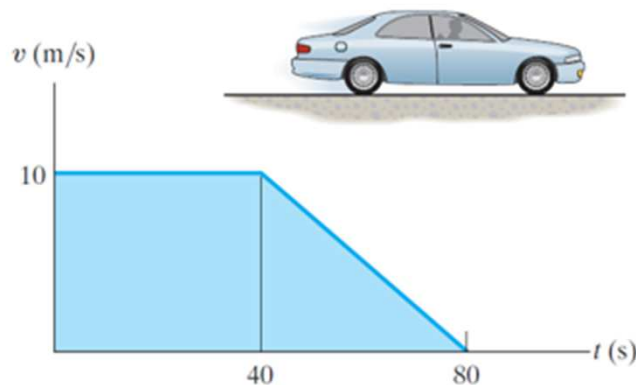
Group Problem Solving I: V-T Graph (1)

Solution:

Find the a-t graph.

For $0 \leq t \leq 40$ $a = \frac{dv}{dt} = 0 \frac{m}{s^2}$

For $40 \leq t \leq 80$ $a = \frac{dv}{dt} = \frac{-10}{40} = -0.25 \frac{m}{s^2}$



Group Problem Solving I: V-T Graph (1)

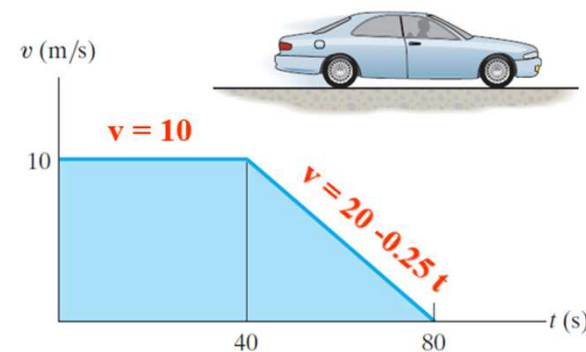
Now find the distance traveled:

$$\Delta s_{0-40} = \int v \, dt = \int 10 \, dt = 10 (40) = 400 \, m$$

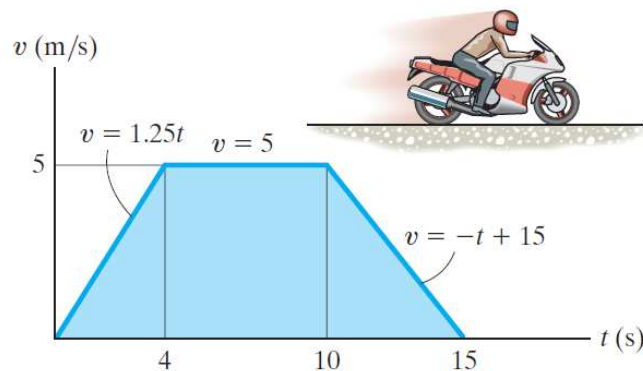
$$\begin{aligned} \Delta s_{40-80} &= \int v \, dt \\ &= \int (20 - 0.25t) \, dt \\ &= [20t - 0.25(1/2)t^2]_{40}^{80} = 200 \, m \end{aligned}$$

$$s_{0-80} = 400 + 200 = 600m$$

$$\begin{aligned} v_{avg(0-80)} &= \text{total distance} / \text{time} \\ &= 600/80 \\ &= 7.5 \, m/s \end{aligned}$$



Group Problem Solving II: V-T Graph (2)



Given: The v-t graph shown.

Find: The a-t graph and distance traveled for the 0 - 15s interval.

Plan: Find slopes of the v-t curve and draw the a-t graph.

Find the area under the curve. It is the distance traveled.



Group Problem Solving II: V-T Graph (2)

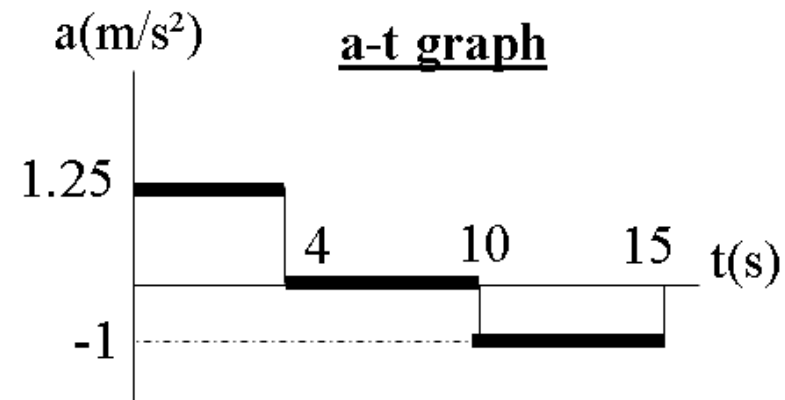
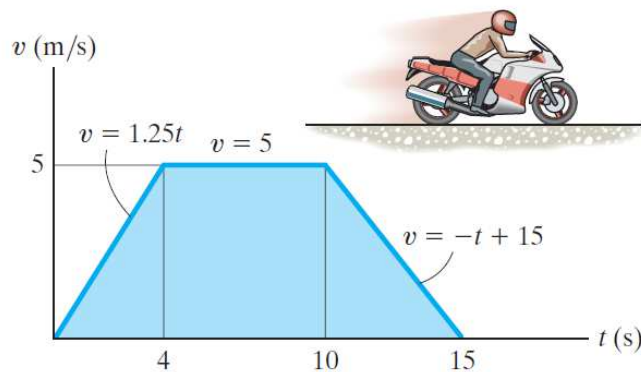
Solution:

Find the a-t graph.

$$\text{For } 0 \leq t \leq 4 \quad a = \frac{dv}{dt} = 1.25 \frac{m}{s^2}$$

$$\text{For } 4 \leq t \leq 10 \quad a = \frac{dv}{dt} = 0 \frac{m}{s^2}$$

$$\text{For } 10 \leq t \leq 15 \quad a = \frac{dv}{dt} = -1 \frac{m}{s^2}$$



Group Problem Solving II: V-T Graph (2)

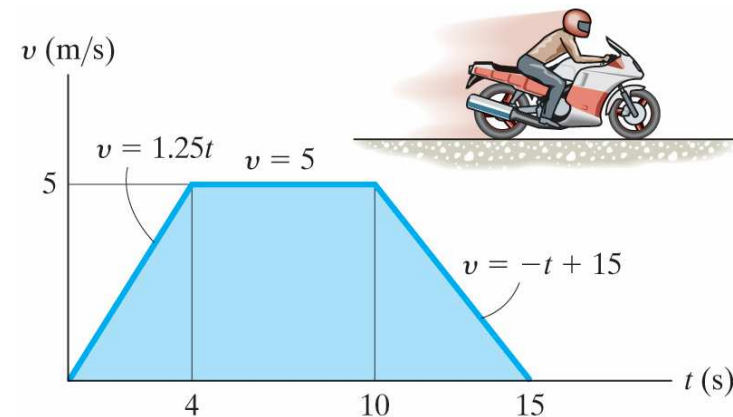
Now find the distance traveled:

$$\Delta s_{0-4} = \int v \, dt = \left[(1.25) (1/2) t^2 \right]_0^4 = 10 \, m$$

$$\Delta s_{4-10} = \int v \, dt = [5t]_4^{10} = 30 \, m$$

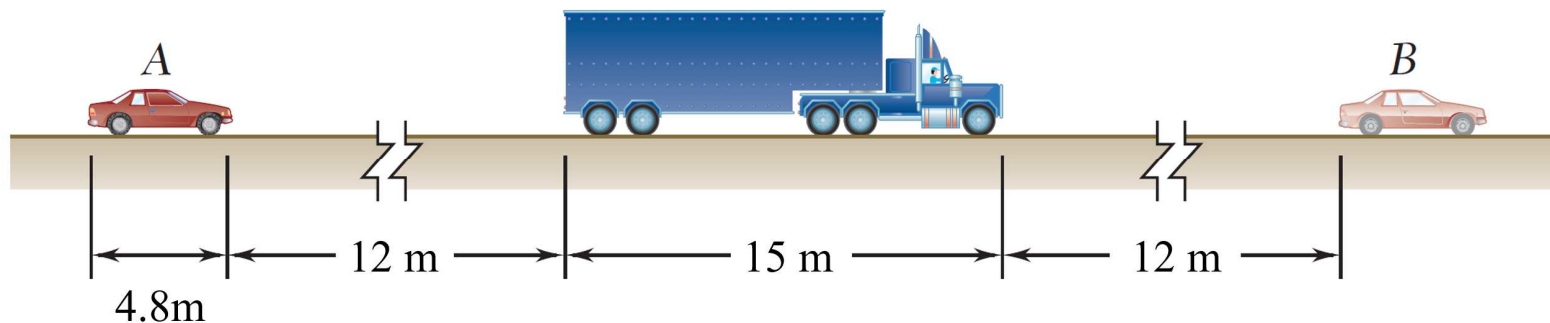
$$\Delta s_{10-15} = \int v \, dt = \left[- (1/2) t^2 + 15 t \right]_{10}^{15} = 12.5 \, m$$

$$s_{0-15} = 10 + 30 + 12.5 = 52.5 \, m$$



Example: Passing Truck Scenario

A car and a truck are both traveling at the constant speed of 56 km/h; the car is 12 m behind the truck. The driver of the car wants to pass the truck, i.e., he wishes to place his car at B, 12 m in front of the truck, and then resume the speed of 56 km/h. The maximum acceleration of the car is 1.5 m/s^2 and the maximum deceleration obtained by applying the brakes is 6 m/s^2 . What is the shortest time in which the driver of the car can complete the passing operation if he does not at any time exceed a speed of 80 km/h? Draw the v-t curve.

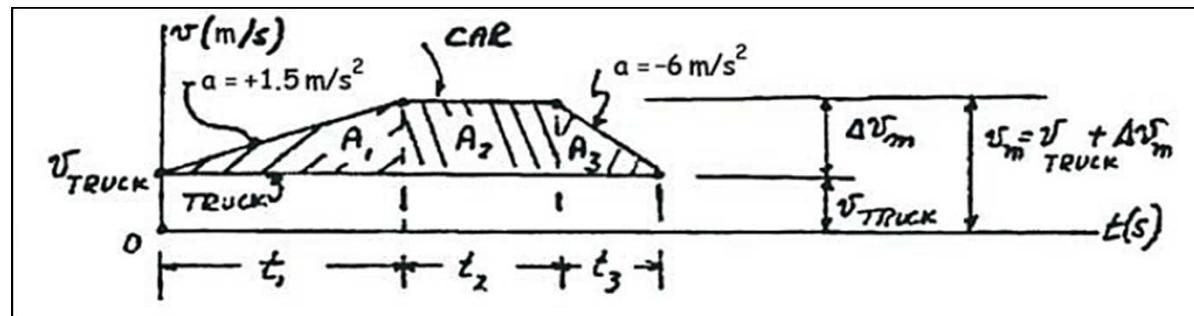


Example: Passing Truck Scenario

Relative to truck, car must move a distance:

$$\Delta x = 4.8 + 12 + 15 + 12 = 43.8 \text{ m}$$

Allowable increase in speed: $\Delta v_m = 80 - 56 = 24 \frac{\text{km}}{\text{h}} = 6.7 \frac{\text{m}}{\text{s}}$



Acceleration Phase: $t_1 = \frac{6.7}{1.5} = 4.5 \text{ s}$

$$A_1 = \frac{1}{2} (6.7)(4.5) = 15.1 \text{ m}$$

Deceleration Phase: $t_3 = \frac{6.7}{6} = 1.1 \text{ s}$

$$A_3 = \frac{1}{2} (6.7)(1.1) = 3.7 \text{ m}$$

$$43.8 \text{ m} = 15.1 + (6.7)t_2 + 3.7$$

$$t_2 = 3.7 \text{ s}$$

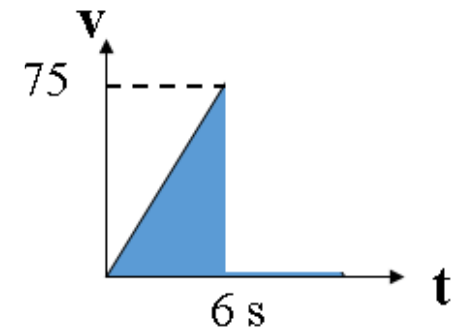
$$t_{\text{total}} = t_1 + t_2 + t_3 = 4.5 + 3.7 + 1.1 = 9.3 \text{ s}$$



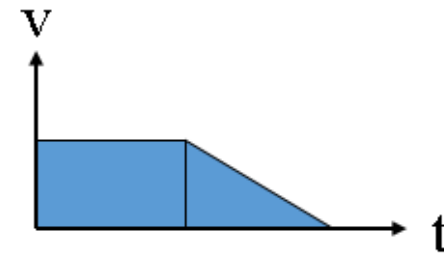
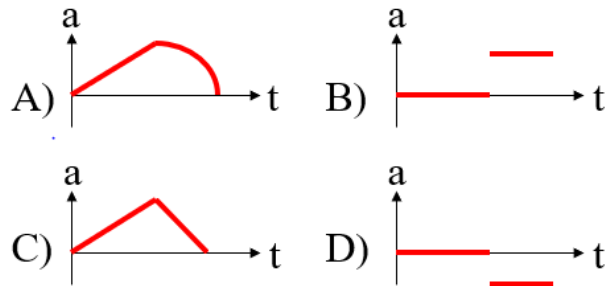
Attention Quiz

1. If a car has the velocity curve shown, determine the time t necessary for the car to travel 100 meters.

- A. 8 s
- B. 4 s
- C. 10 s
- D. 6 s



2. Select the correct a - t graph for the velocity curve shown.



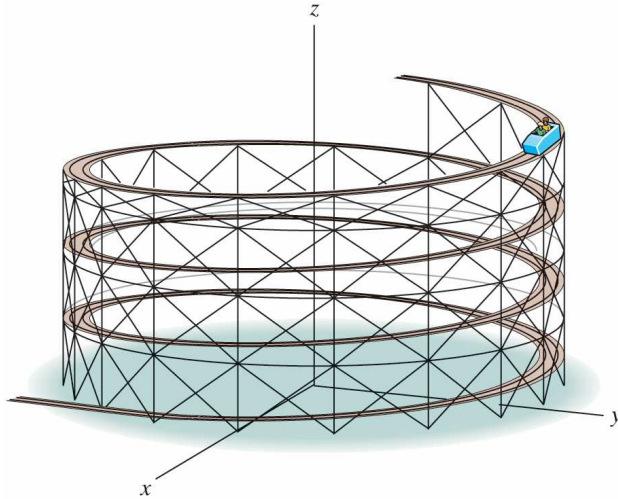
Curvilinear Motion: General & Rectangular Components

The path of motion of a plane can be tracked with radar and its x , y , and z -coordinates (relative to a point on earth) recorded as a function of time.

How can we determine the velocity or acceleration of the plane at any instant?



Applications



A roller coaster car travels down a fixed, helical path at a constant speed.

How can we determine its position or acceleration at any instant?

If you are designing the track, why is it important to be able to predict the acceleration of the car?

The snowboarder and the train both undergo curvilinear motion.

A particle moving along a curve other than a straight line is in curvilinear motion.



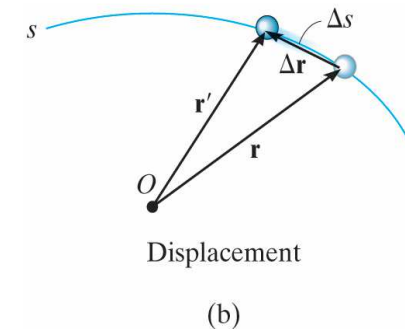
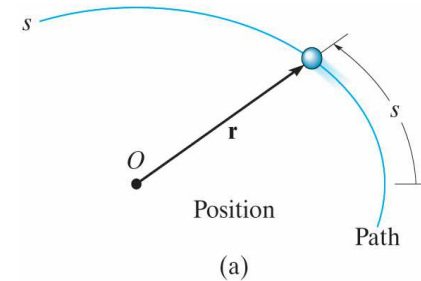
General Curvilinear Motion: Position/Displacement

A particle moving along a curved path undergoes curvilinear motion. Since the motion is often three-dimensional, **vectors** are usually used to describe the motion.

A particle moves along a curve defined by the path function, s .

The **position** of the particle at any instant is designated by the vector $\mathbf{r} = \mathbf{r}(t)$. Both the **magnitude** and **direction** of $\mathbf{r}$ may vary with time.

If the particle moves a distance along the curve during time interval Δt , the displacement is determined by vector subtraction: $\Delta \mathbf{r} = \mathbf{r}' - \mathbf{r}$



General Curvilinear Motion: **Velocity**

Velocity represents the rate of change in the position of a particle.

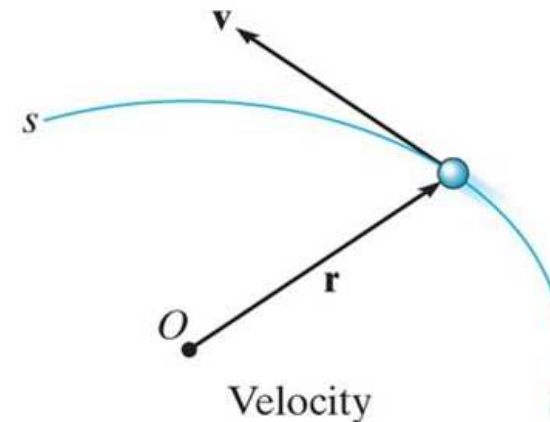
The **average velocity** of the particle during the time increment Δt is

$$v_{avg} = \Delta r / \Delta t .$$

The **instantaneous velocity** is the time-derivative of position $\mathbf{v} = \frac{d\mathbf{r}}{dt}$

The **velocity vector, $\mathbf{v}$** , is **always** tangent to the path of motion.

The magnitude of $\mathbf{v}$ is called the **speed**. Since the arc length Δs approaches the magnitude of Δr as $t \rightarrow 0$, the speed can be obtained by differentiating the path function ($v = \frac{ds}{dt}$). Note that this is not a vector!



General Curvilinear Motion: Acceleration

Acceleration represents the rate of change in the velocity of a particle.

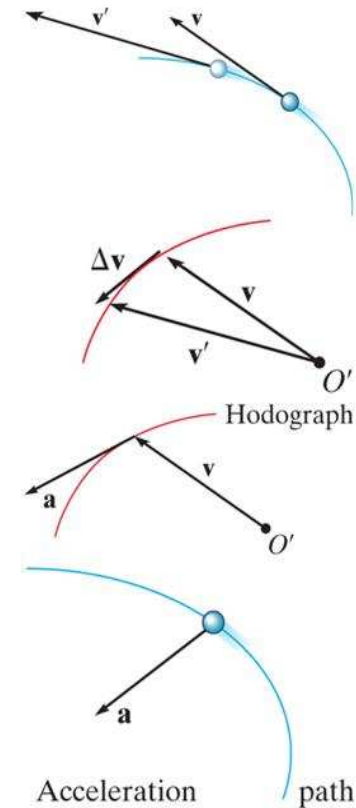
If a particle's velocity changes from $\mathbf{v}$ to $\mathbf{v}'$ over a time increment Δt , the **average acceleration** during that increment is:

$$a_{avg} = \Delta v / \Delta t = (v - v') / \Delta t$$

The **instantaneous acceleration** is the time-derivative of velocity:

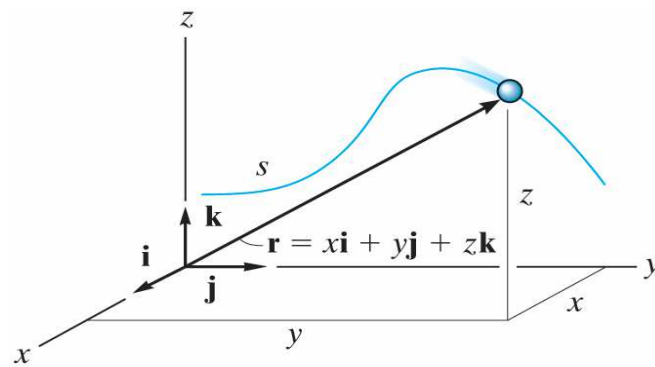
$$a = dv/dt = d^2r/dt^2$$

A plot of the locus of points defined by the arrowhead of the velocity vector is called a **hodograph**. The acceleration vector is tangent to the hodograph, but not, in general, tangent to the path function.



Curvilinear Motion: Rectangular Components

It is often convenient to describe the motion of a particle in terms of its x , y , z or **rectangular components**, relative to a **fixed frame of reference**.



Position

The position of the particle can be defined at any instant by the **position vector**

$$r = xi + yj + zk .$$

The x , y , z -components may all be **functions of time**, i.e.,

$$x = x(t), y = y(t), \text{ and } z = z(t) .$$

The **magnitude** of the position vector is: $r = \sqrt{x^2 + y^2 + z^2}$

The **direction** of r is defined by the unit vector: $u_r = (1/r)r$



Rectangular Components: **Velocity**

The **velocity vector** is the time derivative of the position vector:

$$\mathbf{v} = d\mathbf{r}/dt = d(xi)/dt + d(yj)/dt + d(zk)/dt$$

Since the **unit vectors** $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are **constant** in **magnitude** and **direction**, this equation reduces to

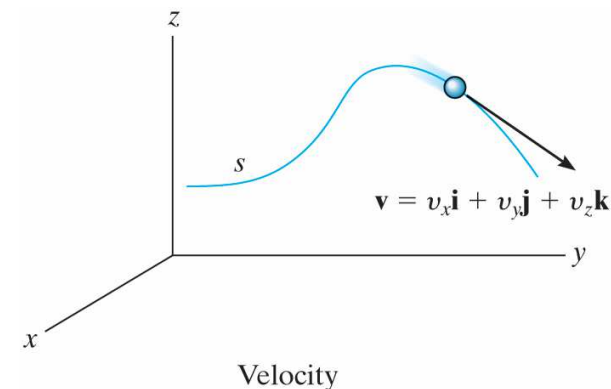
$$\mathbf{v} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$$

where $v_x = \dot{x} = dx/dt$, $v_y = \dot{y} = dy/dt$, $v_z = \dot{z} = dz/dt$

The **magnitude** of the velocity vector is

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

The **direction** of $\mathbf{v}$ is **tangent** to the path of motion.



Rectangular Components: Acceleration

The **acceleration vector** is the time derivative of the velocity vector (second derivative of the position vector).

$$\mathbf{a} = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2 = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

where

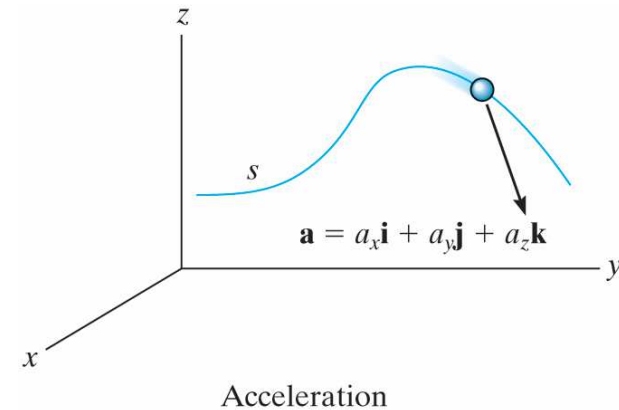
$$a_x = \dot{v}_x = \ddot{x} = dv_x/dt, \quad a_y = \dot{v}_y = \ddot{y} = \frac{dv_y}{dt}$$

$$a_z = \dot{v}_z = \ddot{z} = dv_z/dt$$

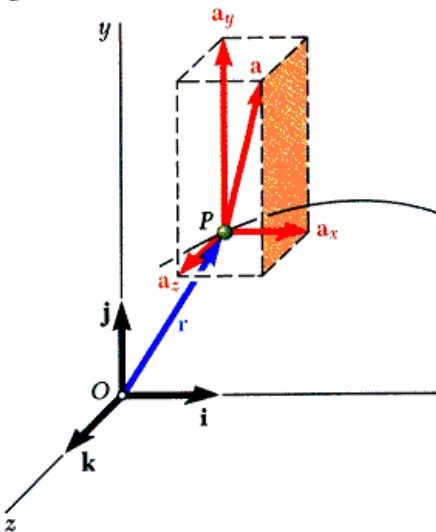
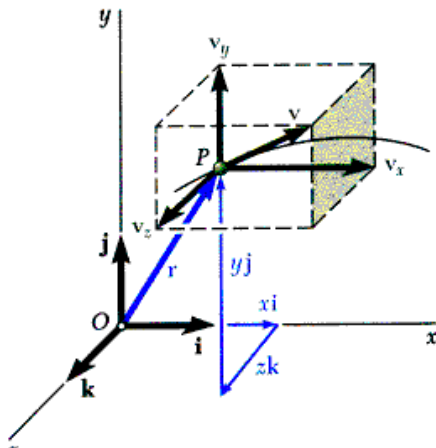
The **magnitude** of the acceleration vector is

$$a = \sqrt{(a_x)^2 + (a_y)^2 + (a_z)^2}$$

The **direction** of $\mathbf{a}$ is **usually not tangent** to the path of the particle.



Rectangular Components Summary



- When position vector of particle P is given by its rectangular components,

$$r = xi + yj + zk .$$

- Velocity vector,

$$v = dr/dt = d(xi)/dt + d(yj)/dt + d(zk)/dt$$

$$v = v_x i + v_y j + v_z k$$

- Acceleration vector,

$$a = dv/dt = d^2r/dt^2 = a_x i + a_y j + a_z k$$



Example: Sliding Box

Given: The box slides down the slope described by the equation

$$y = (0.05x^2) \text{ m}, \quad \text{where } x \text{ is in meters.}$$

$$v_x = -3 \text{ m/s}, a_x = -1.5 \text{ m/s}^2 \text{ at } x = 5 \text{ m.}$$

Find: The y components of the velocity and the acceleration of the box at $x = 5 \text{ m}$.

Plan: Note that the particle's velocity can be found by taking the first time derivative of the path's equation. And the acceleration can be found by taking the second time derivative of the path's equation.

Take a derivative of the position to find the component of the velocity and the acceleration.



Example: Sliding Box

Solution:

Find the y-component of velocity by taking a time derivative of of the position $y = (0.05x^2)$

$$\Rightarrow \dot{y} = 2(0.05)x\dot{x} = 0.1x\dot{x}$$

Find the acceleration component by taking a time derivative of the velocity $\dot{y}$

$$\Rightarrow \ddot{y} = 0.1\dot{x}\dot{x} + 0.1x\ddot{x}$$

Substituting the x-component of the acceleration, velocity at $x=5$ into $\dot{y}$ and $\ddot{y}$



Example: Sliding Box

Since

$$\dot{x} = v_x = -3 \text{ m/s}, \quad \ddot{x} = a_x = -1.5 \text{ m/s}^2 \quad \text{at} \quad x = 5 \text{ m}$$

$$\Rightarrow \ddot{y} = 0.1\dot{x}\dot{x} + 0.1\dot{x}\ddot{x}$$

$$= 0.1(-3)^2 + 0.1(5)(-1.5)$$

$$= 0.9 - 0.75$$

$$= 0.15 \text{ m/s}^2$$

At $x = 5 \text{ m}$

$$v_y = -1.5 \text{ m/s} = 1.5 \text{ m/s} \downarrow$$

$$a_y = 0.15 \text{ m/s}^2 \uparrow$$



Concept Quiz

1. If the position of a particle is defined by $r = [(1.5t^2 + 1)\mathbf{i} + (4t - 1)\mathbf{j}] (m)$, its speed at $t = 1 \text{ s}$ is ____
 - A) 2 m/s
 - B) 3 m/s
 - C) 5 m/s
 - D) 7 m/s
2. The path of a particle is defined by $y = 0.5x^2$. If the component of its velocity along the x-axis at $x = 2 \text{ m}$ is $v_x = 1 \text{ m/s}$ its velocity component along the y-axis at this position is ____.
 - A) 0.25 m/s
 - B) 0.5 m/s
 - C) 1 m/s
 - D) 2 m/s



Group Problem Solving: Particle Movement

Given: The particle travels along the path $y = 0.5 x^2$.

When $t = 0$, $x = y = z = 0$.

Find: The particle's distance and the magnitude of its acceleration when $t = 1$ s, if $v_x = (5 t) \text{ m/s}$, where t is sec.

- Plan:**
- 1) Determine x and a_x by integrating and differentiating v_x , respectively, using the initial conditions.
 - 2) Find the y -component of velocity & acceleration by taking a time derivative of the path.
 - 3) Determine the magnitude of the acceleration & position.



Group Problem Solving: Particle Movement

Solution:

1) x-components:

Velocity known as: $v_x = \dot{x} = (5t) \text{ m/s} \Rightarrow 5 \text{ m/s at } t = 1\text{s}$

Position: $\int v_x dt = \int_0^t (5t) dt \Rightarrow x = 2.5 t^2 \Rightarrow 2.5\text{m at } t = 1\text{s}$

Acceleration: $a_x = \ddot{x} = d/dt(5t) \Rightarrow 5 \text{ m/s}^2 \text{ at } t = 1\text{s}$

2) y-components:

Position known as: $y = 0.5 x^2 \Rightarrow 3.125\text{m at } t = 1\text{s}$

Velocity: $\dot{y} = 0.5(2)x\dot{x} = x\dot{x} \Rightarrow 12.5 \text{ m/s at } t = 1\text{s}$

Acceleration: $a_y = \ddot{y} = \dot{x}\dot{x} + x\ddot{x} \Rightarrow 37.5 \text{ m/s}^2 \text{ at } t = 1\text{s}$



Group Problem Solving: Particle Movement

3) The position vector and the acceleration vector are

Position vector: $r = [xi + yj]m$

where $x = 2.5m, \quad y = 3.125m$

Magnitude: $r = \sqrt{2.5^2 + 3.125^2} = 4.00m$

Acceleration vector: $a = [a_x i + a_y j]m/s^2$

where $a_x = 5m/s^2, a_y = 37.5m/s^2$

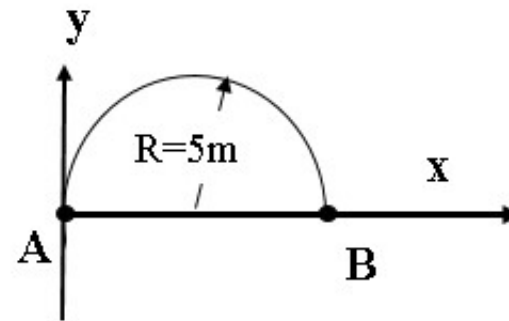
Magnitude: $a = \sqrt{5^2 + 37.5^2} = 37.8m/s^2$



Attention Quiz

- 1) If a particle has moved from A to B along the circular path in 4s, what is the average velocity of the particle?

- A) $2.5\mathbf{i} \text{ m/s}$
- B) $2.5\mathbf{i} + 1.25\mathbf{j} \text{ m/s}$
- C) $1.25\pi\mathbf{i} \text{ m/s}$
- D) $1.25\pi\mathbf{j} \text{ m/s}$

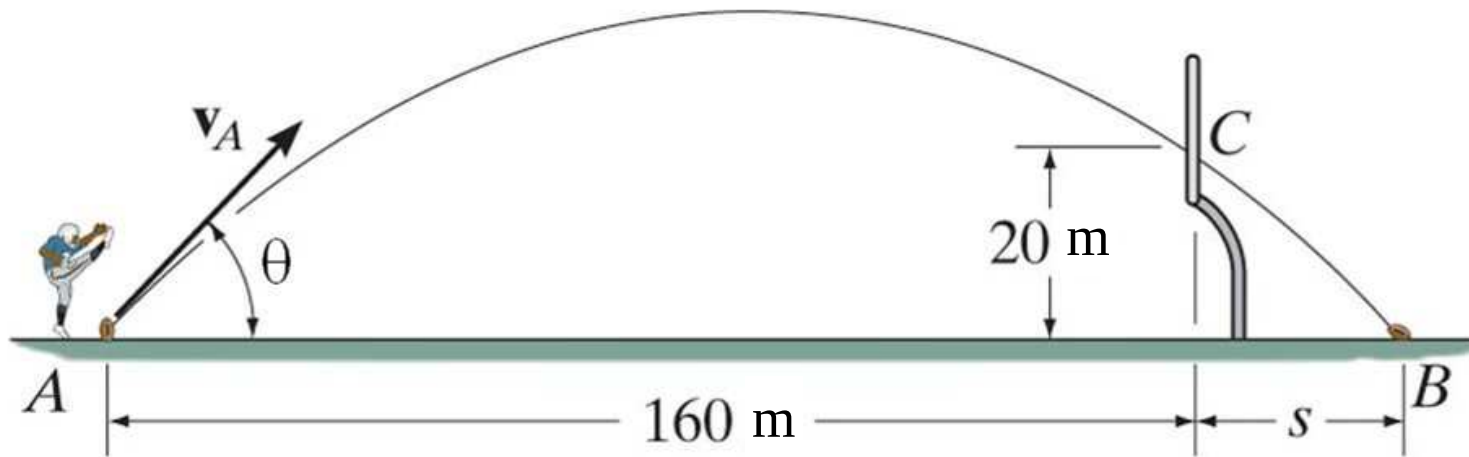


- 2) The position of a particle is given as $\mathbf{r} = (4t^2\mathbf{i} - 2x\mathbf{j}) \text{ m}$. Determine the particle's acceleration.

- A) $(4\mathbf{i} + 8\mathbf{j})\text{m/s}^2$
- B) $(8\mathbf{i} - 16\mathbf{j}) \text{ m/s}^2$
- C) $(8\mathbf{i})\text{m/s}^2$
- D) $(8\mathbf{j})\text{m/s}^2$



Motion of a Projectile



A good kicker instinctively knows at what angle, θ , and initial velocity, v_A , he must kick the ball to make a field goal.

For a given kick “strength”, at what angle should the ball be kicked to get the maximum distance?

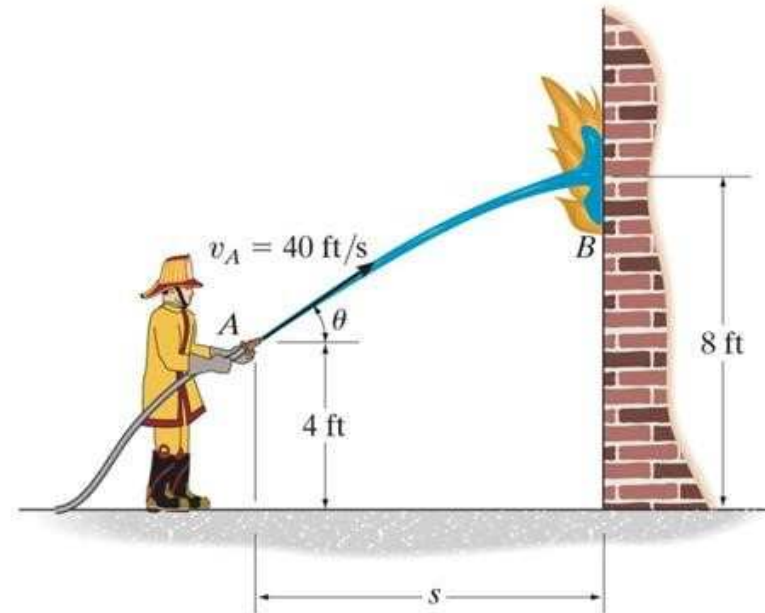


Applications



A basketball is shot at a certain angle. What parameters should the shooter consider in order for the basketball to pass through the basket?

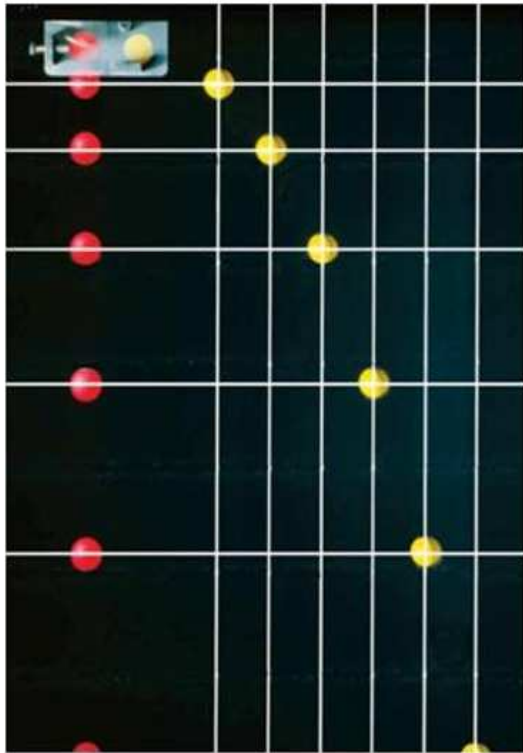
Distance, speed, the basket location, ... anything else?



A firefighter needs to know the maximum height on the wall she can project water from the hose. What parameters would you program into a wrist computer to find the angle, θ that she should use to hold the hose?



Motion of a Projectile

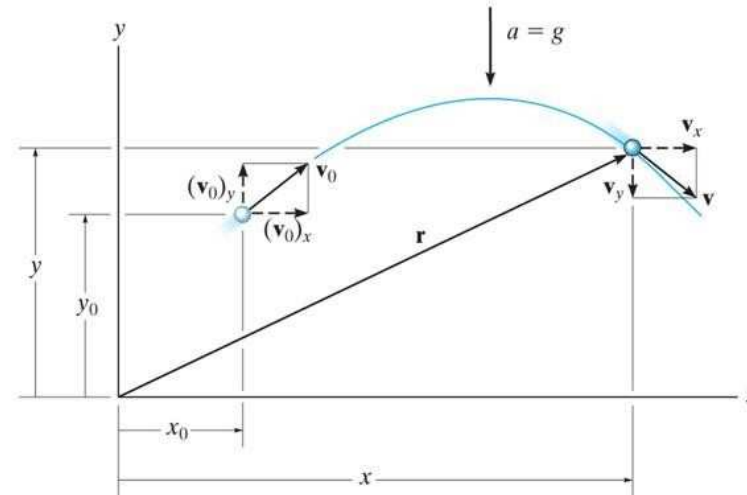


Projectile motion can be treated as two rectilinear motions, one in the horizontal direction experiencing **zero acceleration** and the other in the vertical direction experiencing **constant acceleration** (i.e., from gravity).

For illustration, consider the two balls on the left. The red ball falls from rest, whereas the yellow ball is given a horizontal velocity. Each picture in this sequence is taken after the same time interval. Notice both balls are subjected to the same downward acceleration since they remain at the same elevation at any instant. Also, note that the horizontal distance between successive photos of the yellow ball is constant since the velocity in the horizontal direction is constant.



Kinematic Equations: Horizontal Motion



Since $a_x = 0$, the velocity in the horizontal direction remains constant ($v_x = v_{ox}$) and the position in the x direction can be determined by:

$$x = x_o + (v_{ox}) t$$

Why is a_x equal to zero (what assumption must be made if the movement is through the air)?



Kinematic Equations: Vertical Motion

Since the positive y -axis is directed upward, $a_y = -g$.

Application of the constant acceleration equations yields:

$$v_y = v_{oy} - g t$$

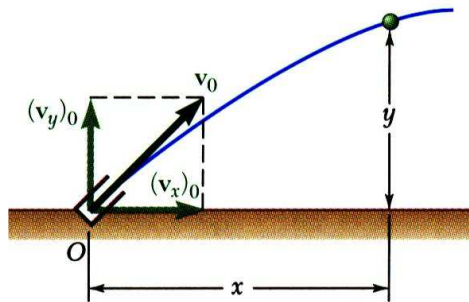
$$y = y_o + (v_{oy}) t - \frac{1}{2} g t^2$$

$$v_y^2 = v_{oy}^2 - 2 g (y - y_o)$$

For any given problem, only two of these three equations can be used. **Why?**



Rectangular Components: Motion of a Projectile



- Rectangular components particularly effective when component accelerations can be integrated independently, e.g., motion of a projectile,

$$a_x = \ddot{x} = 0 \quad a_y = \ddot{y} = -g \quad a_z = \ddot{z} = 0$$

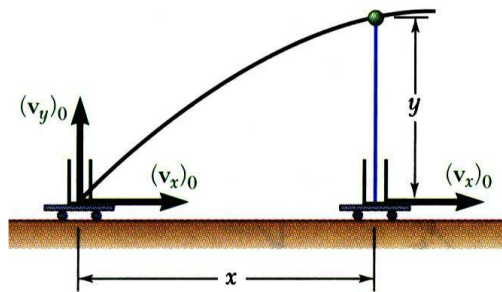
with initial conditions,

$$x_0 = y_0 = z_0 = 0 \quad (v_x)_0, (v_y)_0, (v_z)_0 = 0$$

Integrating twice yields

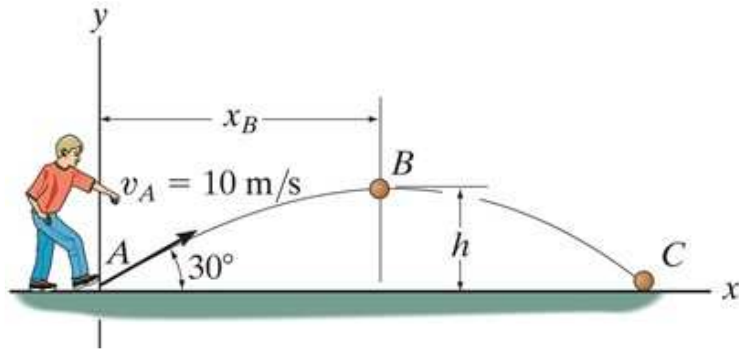
$$v_x = (v_x)_0 \quad v_y = (v_y)_0 - gt \quad v_z = 0$$

$$x = (v_x)_0 t \quad y = (v_y)_0 t - \frac{1}{2}gt^2 \quad z = 0$$



- Motion in horizontal direction is uniform.
- Motion in vertical direction is uniformly accelerated.
- Motion of projectile could be replaced by two independent rectilinear motions.

Example I: Kicking the Ball



Given: v_A and θ

Find: Horizontal distance it travels and v_C .

Plan: Apply the kinematic relations in x - and y -directions.

Solution:

Using $v_{Ax} = 10 \cos 30$ and $v_{Ay} = 10 \sin 30$

We can write $v_x = 10 \cos 30$

$$v_y = 10 \sin 30 - (9.81) t$$

$$x = (10 \cos 30) t$$

$$y = (10 \sin 30) t - \frac{1}{2} (9.81) t^2$$

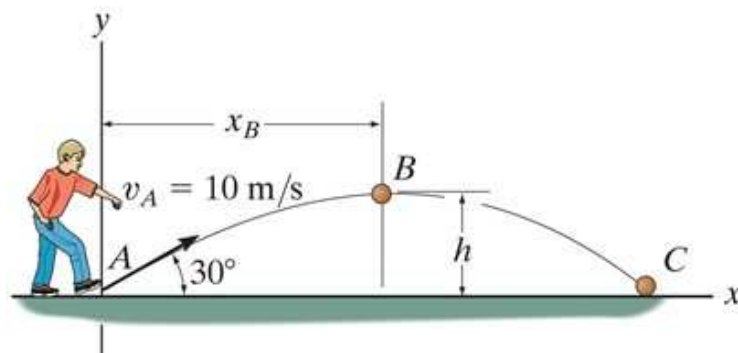
Since $y = 0$ at C, $0 = (10 \sin 30) t - \frac{1}{2} (9.81) t^2 \Rightarrow t = 0,1.019 \text{ s}$



Example I: Kicking the Ball

Only the time of 1.019 s makes sense!

Velocity components at C are;



$$\begin{aligned} v_{Cx} &= 10 \cos 30 \\ &= 8.66 \text{ m/s} \rightarrow \end{aligned}$$

$$\begin{aligned} v_{Cy} &= 10 \sin 30 - (9.81) (1.019) \\ &= -5 \text{ m/s} = 5 \text{ m/s} \downarrow \end{aligned}$$

$$v_C = \sqrt{8.66^2 + (-5)^2} = 10 \text{ m/s}$$

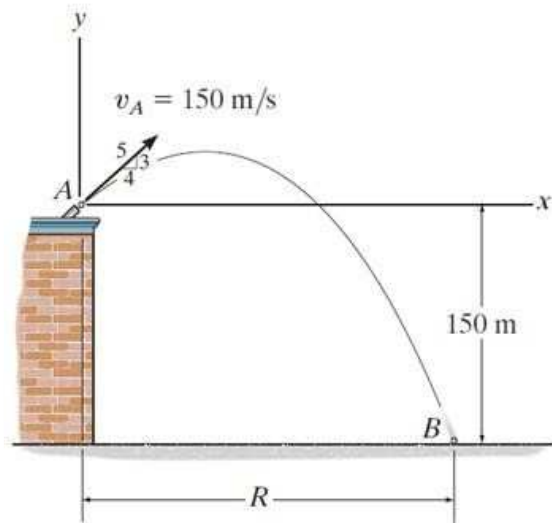
Horizontal distance the ball travels is;

$$x = (10 \cos 30) t$$

$$x = (10 \cos 30) 1.019 = 8.83 \text{ m}$$



Example II: Firing Projectile



Given: Projectile is fired with $v_A = 150 \text{ m/s}$ at point A.

Find: The horizontal distance it travels (R) and the time in the air.

Plan: Establish a fixed x, y coordinate system (in this solution, the origin of the coordinate system is placed at A). Apply the kinematic relations in x - and y - directions.



Example II: Firing Projectile

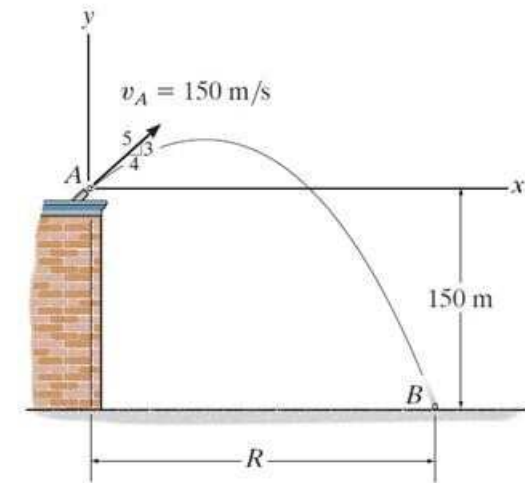
Solution:

- 1) Place the coordinate system at point A. Then, write the **equation for horizontal motion**.

$$+\rightarrow x_B = x_A + v_{Ax} t_{AB}$$

where $x_B = R, x_A = 0, v_{Ax} = 150 (4/5) \text{ m/s}$

Range, R, will be $R = 120 t_{AB}$



- 2) Now write a **vertical motion equation**. Use the distance equation.

$$+\uparrow y_B = y_A + v_{Ay} t_{AB} - 0.5 g t_{AB}^2$$

where $y_B = -150, y_A = 0, \text{ and } v_{Ay} = 150 (3/5) \text{ m/s}$

We get the following equation: $-150 = 90 t_{AB} + 0.5 (-9.81) t_{AB}^2$

Solving for t_{AB} first, $t_{AB} = 19.89 \text{ s}$.

Then, $R = 120 t_{AB} = 120 (19.89) = 2387 \text{ m}$



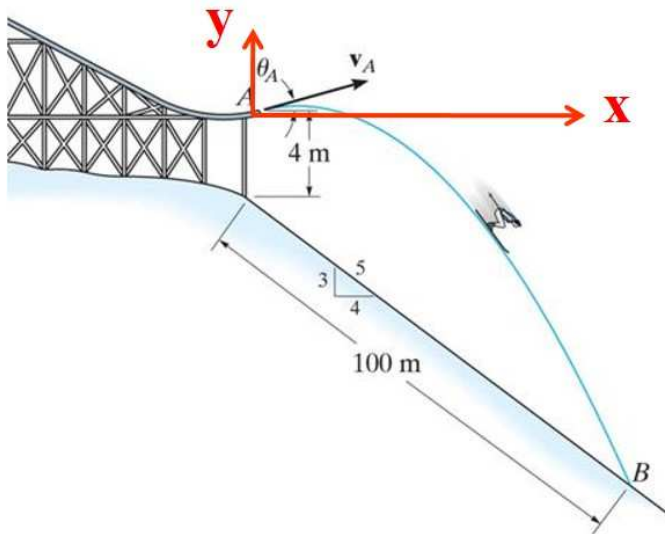
Concept Quiz

1. In a projectile motion problem, what is the maximum number of unknowns that can be solved?
 - A. 1
 - B. 2
 - C. 3
 - D. 4

2. The time of flight of a projectile, fired over level ground, with initial velocity V_o at angle θ , is equal to?
 - A. $(v_o \sin \theta)/g$
 - B. $(2v_o \sin \theta)/g$
 - C. $(v_o \cos \theta)/g$
 - D. $(2v_o \cos \theta)/g$



Group Problem Solving: Ski Jump



Given: A skier leaves the ski jump ramp at $\theta_A = 25^\circ$ and hits the slope at B.

Find: The skier's initial speed v_A .

Plan: Establish a fixed x, y coordinate system (in this solution, the origin of the coordinate system is placed at A). Apply the kinematic relations in x- and y-directions.



Group Problem Solving: Ski Jump

Solution:

Motion in x-direction:

Using $x_B = x_A + v_{ox}(t_{AB}) \Rightarrow (4/5)100 = 0 + v_A(\cos 25^\circ)t_{AB}$

$$t_{AB} = \frac{80}{v_A (\cos 25^\circ)} = \frac{88.27}{v_A}$$

Motion in y-direction:

Using $y_B = y_A + v_{oy}(t_{AB}) - \frac{1}{2} g(t_{AB})^2$

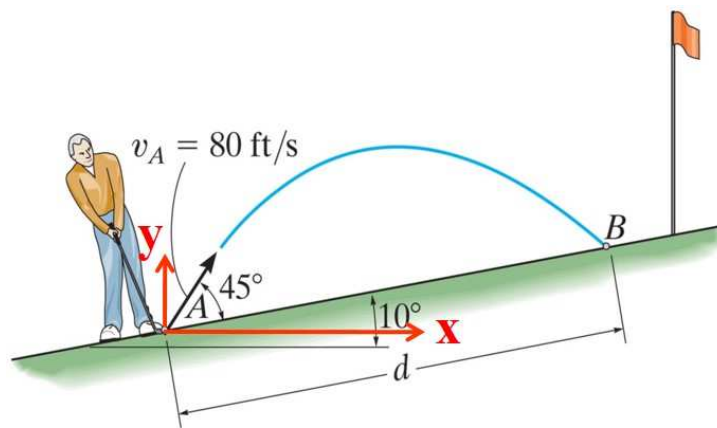
$$-64 = 0 + v_A(\sin 25^\circ) \left\{ \frac{88.27}{v_A} \right\} - \frac{1}{2} (9.81) \left\{ \frac{88.27}{v_A} \right\}^2$$

$$v_A = 19.42 \text{ m/s}$$

$$t_{AB} = (88.27 / 19.42) = 4.54 \text{ s}$$



Group Problem Solving: Golf Ball Trajectory



Given: The golf ball is struck with a velocity of 80 ft/s as shown.

Find: Distance d to where it will land.

Plan: Establish a fixed x, y coordinate system (in this solution, the origin of the coordinate system is placed at A). Apply the kinematic relations in x - and y -directions.



Group Problem Solving: Golf Ball Trajectory

Solution:

Motion in x-direction:

$$\text{Using } x_B = x_A + v_{ox}(t_{AB})$$

$$\Rightarrow d \cos 10 = 0 + 80 (\cos 55) t_{AB}$$

$$t_{AB} = 0.02146 d$$

Motion in y-direction:

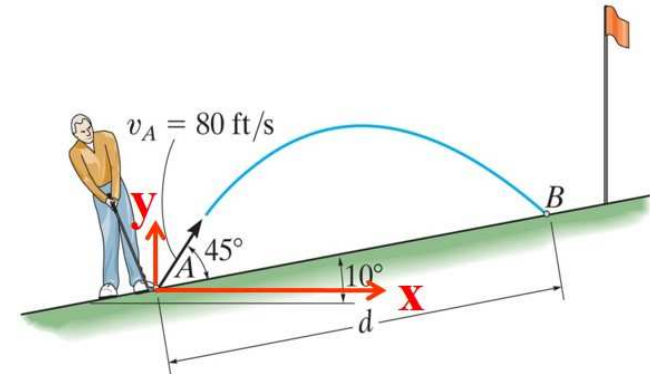
$$\text{Using } y_B = y_A + v_{oy}(t_{AB}) - \frac{1}{2}g(t_{AB})^2$$

$$\Rightarrow d \sin 10 = 0 + 80(\sin 55)(0.02146 d) - \frac{1}{2} 32.2 (0.02146 d)^2$$

$$\Rightarrow 0 = 1.233 d - 0.007415 d^2$$

$$d = 0, 166 ft$$

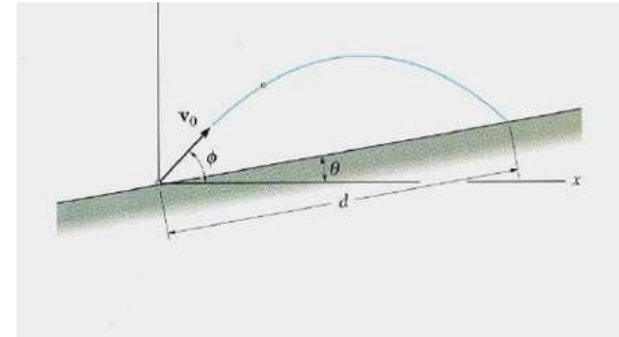
Only the non-zero answer is meaningful.



Attention Quiz

1. A projectile is given an initial velocity v_o at an angle ϕ above the horizontal. The velocity of the projectile when it hits the slope is _____ the initial velocity

- A. less than
- B. equal to
- C. greater than
- D. None of the above.



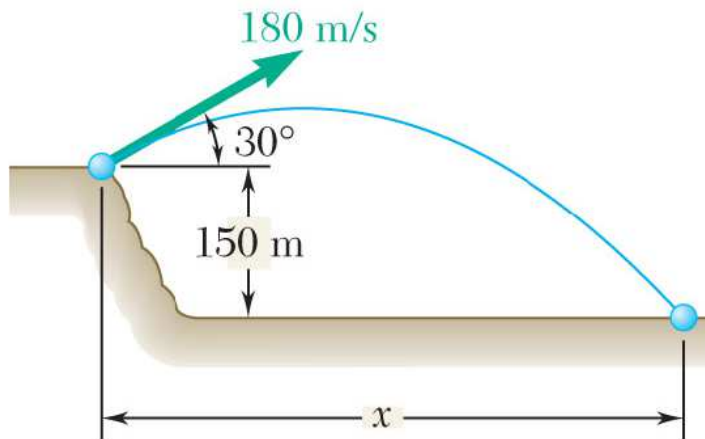
2. A particle has an initial velocity v_o at an angle ϕ with respect to the horizontal. The maximum height it can reach is when

- | | |
|----------------------|----------------------|
| A. $\phi = 30^\circ$ | B. $\phi = 45^\circ$ |
| C. $\phi = 60^\circ$ | D. $\phi = 90^\circ$ |



Sample Problem: Projectile Motion

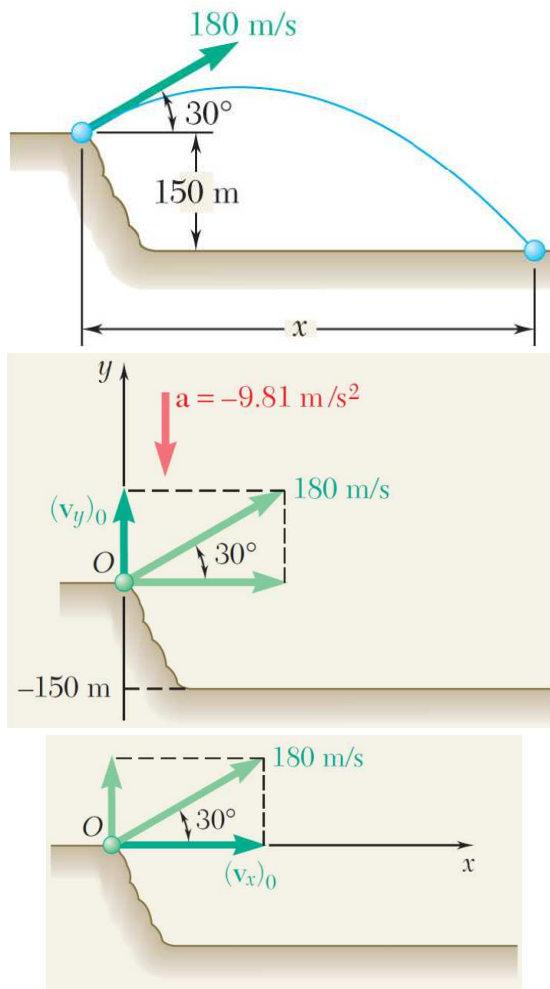
A projectile is fired from the edge of a 150-m cliff with an initial velocity of 180 m/s at an angle of 30° with the horizontal. Neglecting air resistance, find (a) the horizontal distance from the gun to the point where the projectile strikes the ground, (b) the greatest elevation above the ground reached by the projectile.



STRATEGY:

- Consider the vertical and horizontal motion separately (they are independent)
- Apply equations of motion in y-direction
- Apply equations of motion in x-direction
- Determine time t for projectile to hit the ground, use this to find the horizontal distance
- Maximum elevation occurs when $v_y = 0$

Sample Problem: Projectile Motion



MODELING and ANALYSIS:

Given: $(v)_0 = 180 \text{ m/s}$ $(y)_0 = 150 \text{ m}$
 $(a)_y = -9.81 \text{ m/s}^2$ $(a)_x = 0 \text{ m/s}^2$

Vertical motion – uniformly accelerated:

$$(v_y)_0 = (180 \text{ m/s}) \sin 30^\circ = +90 \text{ m/s}$$

$$v_y = (v_y)_0 + at \quad v_y = 90 - 9.81t \quad (1)$$

$$y = (v_y)_0 t + \frac{1}{2}at^2 \quad y = 90t - 4.90t^2 \quad (2)$$

$$v_y^2 = (v_y)_0^2 + 2ay \quad v_y^2 = 8100 - 19.62y \quad (3)$$

Horizontal motion – uniformly accelerated:

Choose positive x to the right as shown

$$(v_x)_0 = (180 \text{ m/s}) \cos 30^\circ = +155.9 \text{ m/s}$$

$$x = (v_x)_0 t \quad x = 155.9t$$

Sample Problem: Projectile Motion

MODELING and ANALYSIS:

Horizontal distance

Projectile strikes the ground at: $y = -150$ m

Substitute into equation (1) above

$$-150 = 90t - 4.90t^2$$

Solving for t , we take the positive root

$$t^2 - 18.37t - 30.6 = 0 \quad t = 19.91 \text{ s}$$

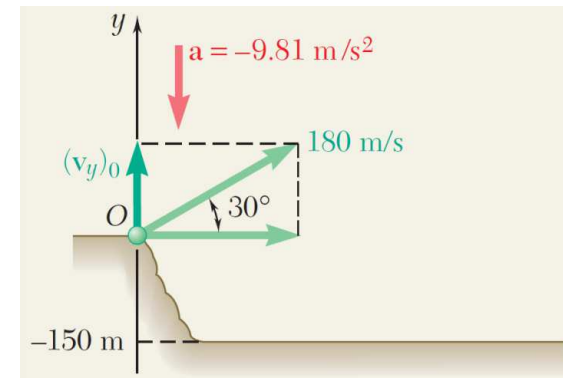
Substitute t into equation (4)

$$x = 155.9(19.91) \quad x = 3100 \text{ m}$$

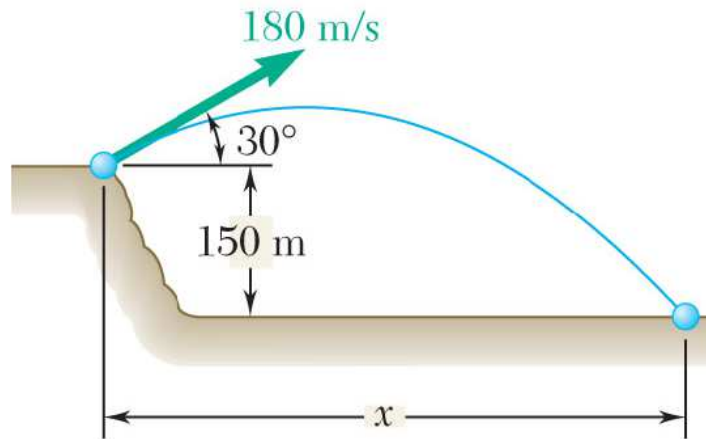
Maximum elevation occurs when $v_y = 0$

$$0 = 8100 - 19.62y \quad y = 413 \text{ m}$$

$$\text{Maximum elevation above the ground} = 150 \text{ m} + 413 \text{ m} = 563 \text{ m}$$



Sample Problem: Projectile Motion



REFLECT and THINK:

Because there is no air resistance, you can treat the vertical and horizontal motions separately and can immediately write down the algebraic equations of motion. If you did want to include air resistance, you must know the acceleration as a function of speed, and then you need to use the basic kinematic relationships, separate variables, and integrate.

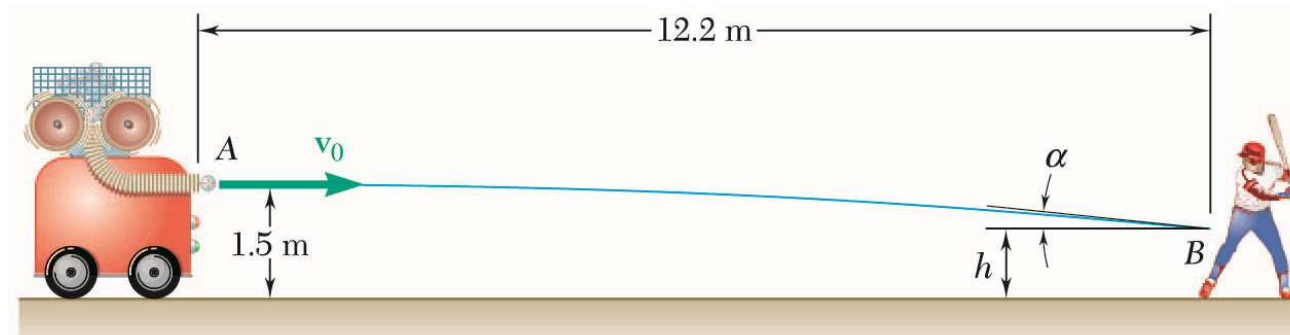


Group Problem Solving: Pitching Machine

A baseball pitching machine “throws” baseballs with a horizontal velocity v_0 . If you want the height h to be 1050 mm, determine the value of v_0 .

STRATEGY:

- Consider the vertical and horizontal motion separately (they are independent)
- Apply equations of motion in y-direction
- Apply equations of motion in x-direction
- Determine time t for projectile to fall to 1050 mm
- Calculate v_0

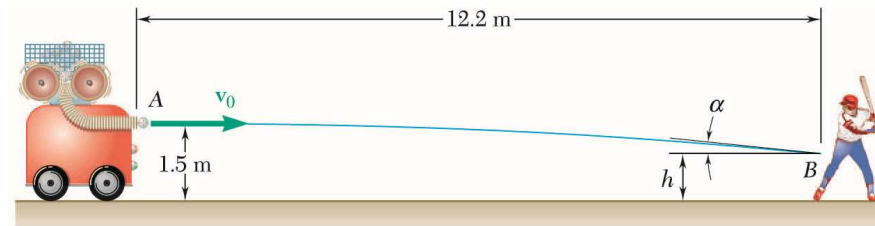


Group Problem Solving: Pitching Machine

MODELING and ANALYSIS:

Given: $x = 12.2 \text{ m}$, $y_0 = 1.5 \text{ m}$
 $y_f = 1050 \text{ mm}$.

Find: v_0



Analyze the motion in the y-direction

$$y_f = y_0 + (0)t - \frac{1}{2}gt^2$$

$$1.05 = 1.5 - \frac{1}{2}gt^2$$

$$0.45 \text{ m} = -\frac{1}{2}(9.81 \text{ m/s}^2)t^2$$

$$t = 0.30289 \text{ s}$$

Analyze the motion in the x-direction

$$x = 0 + (v_x)_0 t = v_0 t$$

$$12.2 \text{ m} = (v_0)(0.30289 \text{ s})$$

$$v_0 = 40.3 \text{ m/s} = 145 \text{ km/h}$$

REFLECT and THINK:

Units are correct and magnitudes are reasonable

